Evaluating Plausibility in VR Beach Volleyball with varying levels of Crowd Population

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Supervisor: Dr. Gareth Young

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Declaration

I, the undersigned, declare that this work has not previously been submitted as an exercise for a degree at this, or any other University, and that unless otherwise stated, is my own work.

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...ABSTRACT...

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Chapter 1

Introduction

Athletes in professional, highly competitive sports have every interest in resorting to the newest advances in technology in order to enhance their competitive performance, and training regime. With recent jumps in technology, Virtual Reality (VR) is increasingly establishing itself as a valuable tool to professional athletes in training scenarios, thanks to its ability to display a wider variety of settings and activity, at a higher level of control than training grounds ever could grant on a regular basis.

Earliest adoptions of VR, or immersive technologies in professional sport include 3D recreations of matches for more immersive match reviews from either first-person, or 3-dimensional non-fixed viewpoints in the early 2010s (Farley et al. (2020)).
The 2022 inews article "Half of Premier League clubs use virtual reality to heal injuries and recreate matchday pressure in training" (Cunningham (2022)) highlights the increasing popularity of VR across the elite-level football, or soccer, scene, with half of the twenty English Premier League teams of the time applying VR scenarios for injury rehabilitation, recreation of matchday atmosphere and pressures, and other training and academy uses, which were left more vague.
The advantages of this technology are manifold: besides the proven ability to positively affect performance and motivation through various different factors interacting together, e.g. immersion (Ijsselsteijn et al. (2004)), competitiveness (Nunes et al. (2014)), digital cooperation (Irwin et al. (2012)), agency in selecting and adjusting the environment (Neumann et al. (2018)), or adaptive difficulty levels (Gray (2017), Richlan et al. (2023b)), it is a technology that allows for safer training conditions, examplified with Ritter et al. (2023) showcasing that gymnasts practising on a beam at floor level - determined to be a safer training environment for the study than the physical facility featuring a beam raised above the floor - whilst wearing a Head-Mounted Display (HMD) experienced similar psychological conditions to the control group practising in the physical environment. VR is

today also applicable at a small scale, e.g. in local gyms, or in dedicated spaces at home or in training centres, and affordably through commercial-grade HMDs (Farley et al. (2020)).

With the increased interest, there is a rise in scrutiny of which ways the incorporation of VR can benefit athletes, with potential for use in sharpening motor skills such as movement and coordination, cognitive skills like perception on the pitch or positioning, mental skills to better resist match or score pressure, or tactical and strategic proficiency in helping to understand and apply directives from the coach (Richlan et al. (2023b)). Newer research, such as Ritter et al. (2023) and Gerwann et al. (2025), also promotes the introduction of Augmented Virtuality (AV), which introduces interaction with physical objects to provide or simulate haptic feedback whilst performing tasks in VR.

However, despite the widening adoption of the technology and increased research, the factors affecting performance in, and skill acquisition through VR are still a very young field of research that is yet to accurately study the aspects in a more granular and standardised fashion, as various reviews of existing studies into the topic of performance in VR sports applications, and their effect on competitive output, highlight (Neumann et al. (2018), Richlan et al. (2023b), Gerwann et al. (2025)).

Despite a notable extension of the range of sports and sport tasks studied between the reviews of Neumann et al. (2018) and Richlan et al. (2023b), the former of which advocated for exactly this extension beyond mostly aerobic workout activities, a sport that seems conspicuously absent from studies so far is Volleyball, alongside any of its variations, and probably for good reasons, as it requires dealing with many inherent limitations of VR application to sport, outlined in Richlan et al. (2023b).

That being said, there is ample evidence that the acquisition and refinement of complex skills in sport does not require a perfect recreation of the environment, and can be compartmentalised to fit specific purposes at a time, as evidenced by Farley et al. (2020)'s proposal of a system making use of a skateboard with VR environments, to practise surfing in dry conditions for example. Therefore, volleyball seems a suitable candidate for further investigation.

This dissertation therefore aims to stimulate further research into applying VR to volleyball applications by investigating whether, and in what way, simulated crowds meaningfully affect, both positively or negatively, users' sense of immersion or, more specifically, the illusion of plausibility that can make or break this feeling of immersion - itself a primordial factor in promoting the plethora of positive effects made possible by interactive VR.

Background knowledge required to grasp the various aspects of this study will be outlined before a selection and comparison of existing literature on, or related to, the subject in Chapter 2.
The subject of the study, and proposed solution theretowards will be detailed in Chapter 3, while the technical contributions made throughout this project are showcased in Chapter 4.
The proceedings of the study and its results will be looked into in Chapter 5, before closing this dissertation with a critical look at the study and its results, and suggestions for directions research building upon this paper might be worth taking in Chapter 6.

Chapter 2

State of the Art

In order to set this study apart from existing research into VR application to sport, it is fundamental to understand what angle this paper is approaching the question of plausibility and immersion from, as well as the impact it would aim to make if applied or researched further. Following this exposition, some notable studies that come close to the aims and methodology of this one will be introduced, then compared to each other and the aims of this study in the closing stages of this chapter.

2.1 Background

When dissecting the title of this paper, ”Evaluating Plausibility in VR Beach Volleyball with varying levels of Crowd Population”, there are certain terms that require highlighting and explaining, if one is to understand the aims and approach to this paper.
The first such term, in order of appearance, is ”Plausibility”, however it is a term that first requires understanding and clearly defining the next ”contentious” term, VR.

2.1.1 Virtual Reality

The term Virtual Reality describes a system using modern hardware and computer rendering techniques to produce a realistic looking, digital world for multi-sensorial experiences (Farley et al. (2020)), controlled by a tracking system for input, and display system for output (Kulpa et al. (2015)). In the specific use of HMD-controlled VR, sometimes also referred to as immersive Virtual Reality (iVR) (Gerwann et al. (2025)), the viewpoint within the virtual scene is updated based on the position and rotation of the display. Images displayed in such systems are stereoscopic (rendered for both eyes separately) to provide an illusion of depth (Vignais et al. (2015)).

For the remainder of this paper, and to interpret the title of this study from now on, VR will stand in for iVR, as a HMD-controlled system (a Meta Quest 3 device, to be specific) was used for this project.

2.1.2 Immersion, Illusion of Place, and of Plausibility

When defining the experience of being cognitively and emotionally invested in a non-physical scenario, be it relating to a well-written character in a novel, or making reflexive kicks whilst playing FIFA a little too seriously, most people would speak of being "immersed" in their material.
For the longest time, this sense was mostly a feeling of mental tunneling into these materials, that never really tried to actively blend out the physical environment surrounding an immersed reader, moviegoer, or other spectator and participant of various forms of media. With the increasing availability and practicality of iVR technologies, and notably affordable, commercial-grade HMDs, this sense of immersion was taken to a whole new level. VR hardware is able to produce high-quality virtual environments, that, when combined with a HMD, attempts to entirely phase out the visual environment, instead granting a complete vision of this digital space. Complemented with spatial audio technology, and perhaps in the near future, various devices providing haptic feedback, it is a system that aims to hijack the senses into a new reality.

This complete rewriting of immersion for this specific wing of technology has prompted debate around redefining its meaning to suit the newest flavour of the term.
In his paper on "Spirit of Place and Sense of Place in Virtual Realities" (Relph (2007)), geographer Edward Relph translates the terms Spirit of Place, defined as the distinct identity and identifiability of a physical location, and Sense of Place, described as "the ability to grasp and appreciate the distinctive quality of places", in other words the, notably positive, feelings elicited in a person toward a physical place, to more virtual contexts.
While the main point of that paper was to evaluate what a virtual sense and spirit of place sounds like - concluding that, just like real places, this sense and spirit of digital place will result from long-term interplay between world builders and their communities in creating distinct identities to strong places - the author plays around with the word 'presence' as a large factor in making virtual worlds feel more real, when cautioning about how the border between "real and [...] virtual worlds is already porous".

This sense of presence is in turn indirectly picked up in Mel Slater's "Place Illusion and Plausibility can lead to realistic behaviour in immersive virtual environments" (Slater

(2009)), in which the author presents a definition of the terms "immersion", "Place Illusion (PI)", and "Plausibility Illusion (Psi)" to describe the ways in which iVR can "transform place and even self-representation". These definitions were revisited in the same author's "A Separate Reality: An Update on Place Illusion and Plausibility in Virtual Reality" (Slater et al. (2022)), informed by a decade of further research into the subject.

Immersion is described as an inherent quantitative feature of hardware systems representing the capabilities of these systems at guaranteeing an immersive experience through so-called sensorimotor contingencies, referring to actions we carry out to collect information, e.g. turning our head to view the scenery from a different angle (Slater (2009)).
A certain system can be considered 'more immersive' than another if it has, for example, a higher framerate, larger Field of View (FoV), more ways to control the environment e.g. degrees of freedom on the headset, various tracking methods such as hand, eye, or full body tracking, or better tracking accuracy.

On the other hand, PI derives from the sense of presence tangentially mentioned by Relph, which is defined here as the "sense of being there in the environment depicted by the [VR] system". PI builds upon this sense of presence by implying this feeling is present despite the knowledge that it is not a physically anchored place (Slater (2009), Slater et al. (2022)): "It is the strong illusion of being in a place in spite of the sure knowledge that you are not there".
This more intangible definition makes it a subjective measure that is to be evaluated, according to the author, through more indirect assessments involving feedback from questionnaire and behavioural responses to the environment.

Finally, Psi is a novel addition to the subjective sense of immersion referred to as an introduction to this section, which essentially describes the user reacting to virtual events as if they were real and physical. The exact proposed definition is "the illusion that what is apparently happening is really happening[,] even though you know for sure it is not[.]" (Slater (2009), Slater et al. (2022)), i.e. the illusion that digital events such as a ball flying towards the user are happening, and trigger authentic responses, despite the knowledge these events have no consequences on the physical world, in this case that there would be no feeling of contact and pain when the ball does hit the user's face.
Psi is a notion with a lot of nuance, as it involves the virtual environment responding to actions performed by the user, events within it being directed at the user, and meeting the expectations users would have of the same event happening in reality (Slater et al. (2022)), which makes it a very tricky sentiment to reliably capture and assess, and even

harder to provide universally and reliably due to differing, and at times even contradictory expectations around digital events.
As an example of how subjective this measure is, in a study performed to establish how to improve a VR recreation of a Dire Straits concert, participants were asked to reflect on the plausibility of the concert, and some highlighted disturbance at the mix of the crowd (being in one study too male-dominant, and in a second study too female-dominant), the animations of musicians being out of sync with the music being played, or even the absence of smokers amongst the crowd for historical accuracy of the concert being portrayed (Slater et al. (2023)).

In their revision of the definitions for PI and Psi, Slater et.al. state that both illusions being active results in VR users responding realistically to situations and events in VR. There are many more such factors that interplay with PI and Psi to enhance or negate subjective immersion, such as embodiment or the illusion of body ownership, describing the comfort and identifiability of users with their digital avatars (Slater et al. (2022)), or coherence, describing the reasonability, and predictability of environments (Skarbez et al. (2018)), which are however not directly relevant to this study, so will not be detailed any further.

2.1.3 Beach Volleyball

A quick word is needed to introduce volleyball, to give an overview of its main phases of play and certain types of ball interactions, to explain the technical implementation later.

The main distinction between classic, or indoor volleyball and beach volleyball, beyond the obvious change in setting and resulting changes in mobility and outfits, is the team size, which features two teams of six in classic volleyball, but downsizes to teams of two (FIVB (2026)) for its sandy counterpart, and a few minor variations in rules that are not worth going into for concerns of length, but along the gross lines applied to this study, the rules are more or less the same.

A play, or 'rally' is started by a player serving the ball - releasing it, then hitting it with one hand (either 'underarm': from below, upwards, or 'overarm': from above, forwards) to the opposite side of the net directly (FIVB (2026)).

Following a receive, both teams have to endeavour to land the ball on the ground in the opposite court half, or make the opposing team drop the ball anywhere other than the own half. The official, international governing body of Volleyball identifies several valid ways to manipulate the ball while it is in play (FIVB (2026)):

- The **Dig** is a forearm pass performed by aligning a player's arms to form a platform, to bounce the ball off in a desired direction.
- The **Set** is an overhead pass performed with both hands aimed at completely changing the direction of the ball, usually to set up a spike to hit the ball over the other side. It is a particularly sensitive move as it requires very short contact with the ball to be not be ruled as grabbing and penalised, and requires much sensitivity to aim and weight properly.
- The **Spike** is a powerful, one-handed shot aimed at smashing the ball over the net.

Additionally, the ball is allowed to make contact with any part of a player's body, so long as it is neither grabbed, nor held for any length of time.

During a match, the team having won the last point gets the right to serve the next rally, and a point is awarded for every rally, based on where the ball makes contact with any object other than an active player or the net, and the last team to touch the ball before this contact. A team wins a point if either the opposing team last touches a ball that lands out of bounds, or if the ball lands on the ground in the opposing team's court, regardless of last touch (FIVB (2026)).

Beyond the very basic outline, there are many more rules at various levels of nitpicking such as a maximum of three touches per team before the ball has to be touched by the opponents, as well as more ways to interact with the ball that are not formally recognised by the FIVB as named moves, e.g. the 'poke' which involves lightly touching a ball near the net to lob a block, however as they were not included in the technical contributions and could be outlined over an entire book, and I am sure the page count was already daunting to begin with, any other rules are not worth further consideration for the purposes of this report.

Beach Volleyball was chosen for this study due to the initial plan to also incorporate opponents to play against participants, which would, with only three other players, reduce the load, both computational during runtime and for implementation, in employing them. Subjectively, the fact it also accommodates for nicer scenery may have helped sway the decision as well.

2.1.4 Crowd Population

The last concept to introduce is that of Crowd Population.
Crowds form a significant part of matchday pressure, representing the expectations and emotional involvement from outside the pitch being brought directly to athletes and the sporting staff in general. Performance under public scrutiny, be it public speech, artistic or sporting performance, is a widely studied subject under physical conditions, and as mentioned in Chapter 1, is already a factor motivating football teams to adopt VR technology in their training regimes Cunningham (2022).

Sports under COVID lockdown conditions have provided a unique case study to research the impact of crowds, or the lack thereof, on performances in sports that chose to pursue competitive matches behind closed doors, such as many professional football or soccer leagues in Europe, with studies (Leitner et al. (2023), Leitner and Richlan (2021), Richlan et al. (2023a)) aiming to explain various such discrepancies between pre-COVID and mid-COVID seasons. In particular, inspecting changes in home advantage - the perceived performance boost received from supporters at a team's home ground, the impact of the missing crowd on staff 'emotionality' during matches, and the impact of the missing crowd on match officials respectively.

The findings were that the various effects the crowds were perceived to provide before COVID, with no 'control scenario' to prove or disprove the perception - the boost in motivation on home soil, the stoking of emotional reactions in any direction - mostly dissipated, and change was particularly salient in referee behaviour, who described the games as having been less motivating and less focused than matches with full crowds, despite having been easier to referee, and with more compliant and agreeable players and staff (Richlan et al. (2023a)).

As such, the main factors that determine a crowd's impact can be boiled down to their size and density, and their level of involvement or emotional setting.
Therefore, for the purposes of this study, Crowd Population will be defined as the amount and density of virtual spectators to a VR sports event, as well as their level of energy and reactivity to the happenings on the court or playing field.

2.2 "Feasibility of training athletes for high-pressure situations using virtual reality"

In this study Stinson and Bowman (2014), the authors aimed to prove the viability of VR simulations for sport psychology training by subjecting football/soccer goalkeepers to a high-fidelity immersive simulation of a penalty shootout scenario, a particularly stressful match event. The specific goal was to measure whether the simulation could elicit a real sense of anxiety that could, in theory, then be practised away using this same environment.

The simulation environment bases on three independent variables: known anxiety triggers, field of regard - a measure adjacent to FoV describing the degree to which the display surrounds the user, and simulation fidelity.

The researchers concluded that VR sports applications could indeed elicit feelings of anxiety, both physiologically and in test results, although the findings are mitigated by showcasing an overall low level of anxiety despite the rise compared to the baseline scenario, and that the demographic did not present many people with a higher trait anxiety, meaning its generalisability is somewhat debatable, according to the authors.
Additionally, all independent variables were determined to have contributed to the feeling of anxiety, although the influence of field of regard had an adverse effect than expected, with high FoR leading to decreased confidence in lower anxiety-inducing settings w.r.t. the other two parameters.

2.2.1 Plausibility Assessment

The central measure for this study was anxiety, through both quantitative or physiological and qualitative means. This does indirectly measure Psi, considering that if the simulation was experienced as inauthentic, it would not elicit anxious behaviours whatsoever, though the specific measurement 'fails' to keep track of other factors such as the accuracy of ball movements in a simulated space with bent optics, due to the use of a CAVE-like system.

2.2.2 Methodology

A major difference in fundamental approach is the VR system. This study makes use of a CAVE-like display system, which projects the virtual environment onto a cube-shaped room, with 3D glasses employed to preserve the illusion of depth.
Participants therefore have a very different sense of agency in this simulation, as they defended their goal with their full physical body. Behaviour correctness was established

through directional head tracking to establish whether a user reacted fast enough, and in the correct direction to theoretically save the penalty.

Anxiety was measured through physiological measures including heart rate, heart-rate variability, and galvanic skin response, as well as two standardised anxiety tests, STICSA and CSAI-2R.

25 participants completed the study, excluding pilot participants and those that did not finish the study, and were paid for their time alongside a bonus for good performance, thanks to a small price per save, totaling up to US$30. Ages ranged from 22 to 32, and featured 9 female and 15 male participants, with various levels of experience at competitive levels, and in goalkeeping.

2.2.3 Studied Subject

The study aimed to demonstrate the viability of VR simulations for the psychological training of athletes, by showcasing its ability to elicit anxiety in users. This implicitly makes it a study of Presence as a whole, but particularly plausibility as laid out in the Plausibility Assessment section for this study.

The proposed project, in contrast, aims to assess plausibility more broadly to find out points of friction that could cause breaks in presence, as opposed to this proof-of-concept approach with a specific factor in mind.

2.3 "The sentiment of a virtual rock concert"

The paper "The sentiment of a virtual rock concert" (Slater et al. (2023)) builds upon a similar study two years prior, based around the same premise of establishing through participant feedback in what ways a digital recreation of a historic concert, here a 1983 live performance of "Sultans of Swing" by Dire Straits, can be improved upon to maximise enjoyment, or by proxy, the PI and Psi metrics.

While the first study was described as mainly focusing on that idea, this second study was informed by surprising reports and findings in the previous study, which prompted a few adjustments to see in what ways the feedback changed. Notably, swinging the surrounding crowd which was, in the first study, described as too male-dominant to being entirely composed of female avatars. Furthermore some improvements were made with regards to band animation accuracy, and the timing of various dancing and clapping animations, all of which were also mentioned negatively during the first study.

During both studies, participants were sent a link to complete the study from home (taking place during COVID lockdowns), filling out a questionnaire for demographic data, then running the VR simulation on a HMD, which ran for around 10 minutes. They would be placed in the crowd, surrounded by digital avatars interacting with the music. In study 1 their behaviour was fully independent of the participant, in study 2, some participants randomly selected between Look and NoLook, the latter being a control group, would experience the avatars staring back at them for 1-3s if they were looked at.
Following the simulation, participants were then asked to write an essay about their experience, taking into account the factors of the feeling of being at the concert, interactions with the crowd, specifically whether any helped immerse better, or took from the experience, then answering a few questions on a 1-7 Likert scale relating to copresence (social belonging within the crowd), the willingness to dance along with the crowd, and how close the experience felt to a real concert experience.

The findings largely confirmed those made during the first study, with participants now often being thrown off by being around too many women, both for the occasion of an 80s rock concert and in general, the staring being off-putting and even creepy at times, and still featuring some mild resistance at the band when looked at more closely. However, the reports in this second study seemed to highlight more simple discomfort than the reports of creepy and menacing male audience members. Participants also widely seemed to express a high willingness to dance along with the crowd, and the overall feedback reads a lot like "it's fine until you look at the details", resulting in the assessment that overall Psi is fairly high, and could be improved even more by featuring a more diverse crowd, that stares less, and with more eye for detail regarding things like timing for clapping and musical animation accuracy.

2.3.1 Plausibility Assessment

This study aims to directly estimate Psi levels in participants in order to confirm whether it elicits a reaction at all, that could be construed as a successful implementation, and the sentiment analysis employed as its main methodology for measuring is also used to verify whether this reaction is mostly positive or negative. Whilst the findings are used to improve the simulation itself, the conclusion of the second study is that, for this kind of applications that require a high level of positive Plausibility to be successful, developers should involve users in the design to incrementally improve the experience, as this set of studies has done.

2.3.2 Methodology

In order to assess participants' experience, the authors have asked them to answer a form for demographic data before the experience, then write an essay of at least 150 words relating to their experience, before rating three factors describing specific parts of the experience on a Likert scale.

The goal of that study is similar in nature to the proposed study, namely to assess users' plausibility with the underlying purpose to improve on a simulated environment. The study also features a larger participant pool (n=26) than that both envisioned and actually assessed for this study (n=14), a participant count that is neither unsubstantial, nor so rich that the findings can be considered as widely generalisable as can be, although the participant pool is recruited through more professional channels than the researchers' entourages, seeing as participation was proposed to students in foreign programmes, and some influx was recorded from social media advertisements as well.

The sentiment analysis, particularly using and comparing several different implementations of it, seems a fitting method for the overall reaction, and yields many avenues for comparison and details as evidenced by this section being the largest by far in the report. It allowed the researchers to narrow down participants in clusters of experience, determining there to have been four distinct groups of experience, by the resulting essays.

2.3.3 Studied Subject

The largest difference to the proposed study is the subject of this related work, which was to improve subjective immersion in a virtual concert, meaning focus both in analysis and development was mainly posed on realism and comfort to make the experience as real as possible, when exploring avenues to improve on the simulation. This proposed study aims to explore Plausibility as a factor that is essential to applications in competitive sport, to avoid breaks in immersion that might take athletes out. Same studied metric but to a very different end.

There is some overlap to be found however in the fact the virtual audience was subject to much scrutiny throughout the study e.g. in highlighting the anxieties around genderdominance in both ways, though a dampening to this similarity has to be mentioned due to participants also being cast as spectators, rather than being the spectated ones, as they will be in this study.

2.4 "Indifferent or Enthusiastic? Virtual Audiences Animation and Perception in Virtual Reality"

This study led by a research team around Yann Glémarec looks into configuring a virtual crowd of spectators attending a presentation or public speech made by a VR user, with the goal of providing authentic non-verbal behaviour in the virtual crowd for a given level of enthusiasm for the speaker or subject (Arousal), and attitude towards the subject or speaker (Valence).

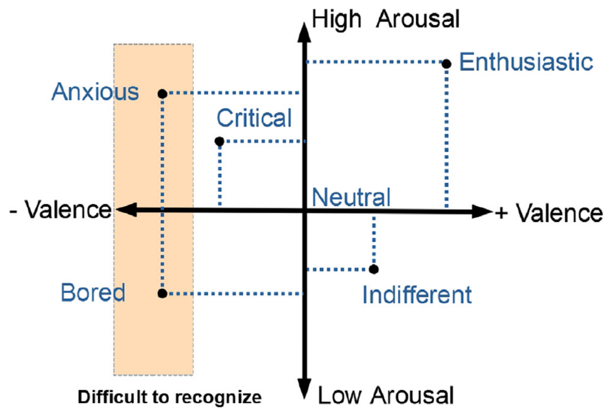


Figure 2.1: A graph showcasing the 2-dimensional scale used in the paper Glémarec et al. (2021) to showcase the relationship between arousal and valence in crowds. The points indicate named configurations used in phase 2 of the study, and the gray area represents an area participants had difficulty differentiating.

The study was conducted in two separate phases: in the first, twenty participants were asked to adjust a crowd's collective posture, specifically gaze direction, the frequency of looking away from this direction, posture openness and proximity towards the speaker, types of head movements and facial expression, and their frequency. At the end of this adjustment game, participants were asked to rate their satisfaction with the result on a 1-7 satisfaction scale.

In the second phase of the study, 38 participants, none of which participated in the first phase of the study, were asked to hold a presentation in front of an audience generated with behaviour extrapolated from the results of the first phase of the study, and asked to reconstruct the Arousal and Valence values associated with the exhibited behaviour. The crowds consisted of 10 avatars, 5 male, 5 female.

The motivation for this study was mostly therapeutic, in helping users become familiar with public speech, and overcome any anxieties or other stresses related to this situation.

The findings of the second study were that crowds with a higher valence, i.e. a more positive view of the speaker or presented topic, were better identified along the arousal scale, meaning that participants were well able to tell an enthusiastic crowd apart from an indifferent one. However there was more difficulty in differentiating arousal levels in crowds with negative valence, meaning that participants found it harder to differentiate between a bored and anxious crowd, each being descriptors at opposite ends of the neutral arousal values, as shown in figure 2.1.
In the guidelines made for audience simulation following the exposition of direct findings, the authors note that the main factors of interpretation for crowd sentiment was the gaze, both the direction and the frequency of change, head movements such as nodding or shaking the head, and the proximity towards the main actor, while noting that facial expressions should alternate between a neutral and a targeted expression to help impress the target, as a sustained facial expression might be interpreted as neutral otherwise, and would predictably be picked up on less if the spectator is further away.
An interesting note in the guidelines was also the authors reinforcing a finding from a different paper based on the results of phase 1, stating that a smaller concentration of negative actors could sway the overall sentiment to negative, stating that the sentiment could be considered positive if only 3 or 4 out of the 10 spectators acted negatively, whilst conversely it needed more than a majority of positive spectators for the sentiment to pass as positive.

2.4.1 Plausibility Assessment

Plausibility or immersion as a whole was not a direct measure of this study, though one could argue it may have been implicitly recorded with the rating of the crowds, since a user that does not believe the crowd to be realistic would not rate it appropriately.

2.4.2 Methodology

The results from the studies here were mostly directly taken from gameplay. In phase 1 of the study, the feature represented the vector of body movements embodied by the virtual crowd, as configured by the participant, paired with the associated valence and arousal values it is to represent. In phase 2, the result is a pair of control V-A pair set by the system, or the 'real' V-A value, and the rating given by the participant, evaluated to compare the disparity.

This methodology showcases the efficacy of letting participants control the configurations to preference, as the evaluation of the configurations was left to be conducted by more participants, instead of imposing any interpretation of crowd sentiment by the research team, as stated by Murcia-Lopez et al. (2020), who conducted a larger study to assess various methodologies to assess plausibility, as a strong point towards this methodology.

2.4.3 Studied Subject

While this subject dives into assessing spectators as the main variable, the conducted study aims more to establish a reliable and recognisable crowd controller, that can accurately recreate the overall sentiment of a set of spectators. It also mainly regards their application in more professional or academic environments, however their findings may possibly be used for more types of crowds, including in sports, though the differences may warrant some dedicated study.

2.5 "Factors Affecting Sense of Presence in a Virtual Reality Social Environment: A Qualitative Study"

In this study aiming to inspect the factors affecting what the authors defined as Sense of Presence, a notion combining Slater's PI and Psi illusions under one hood, Riches et al. (2019) invited 76 participants to participate in a social VR experiment, where they joined a room of digital avatars having a party with the instruction to make an opinion of what the avatars thought of them. Following the scripted VR scenario lasting around 5 minutes, participants then took part in a semi-structured interview to extract the general feel of the experience.

Participants were recruited from a previous study based on scores in the paranoia trait on a Paranoid Thought Scales test, either scoring particularly high or low, and subject to other conditions such as not having a diagnosed history of mental health conditions, or other neurological disorders. 76 participants participated in total, 49 of which were women, and with a mean age of 31.45 years.

The authors found that Presence was strongest when "the VR elicited genuine cognitive, emotional, and behavioral responses; and when participants created their own narrative about events [...]", i.e. when participants described a high sense of Psi. Inversely, Presence was lowest when participants "experienced diminished agency and physical impediments, such as cyber-sickness and awareness of apparatus and body movement".

In terms of the factors that affect Presence, the qualitative analysis of the interviews produced a list of themes that have had an impact on participants' presence, including but not limited to introspective thoughts and emotions, outwards-focused thoughts and emotions, physiological reactions, avatar behaviour, interactivity with the environment, and environmental characteristics, each with various subthemes that interacted differently with participants' presence.

2.5.1 Plausibility Assessment

In this study, the main subject of study was this notion of Presence, defined as a state of consciousness reliant on the perception of 'being there' in VR environments and as 'a psychological state in which virtual objects are experienced as actual objects in either sensory or nonsensory ways'. This definition mirrors quite accurately the combined definition of PI, the sense of "being there" [despite the sure knowledge it is not really the case], and Psi, the illusion that digital events are really happening.

By extension this means it indirectly does measure the cognitive effects of VR, but also incorporates PI in its assessment. As hinted in the description of the findings above, the observation was made that Psi had the greater positive effect on Presence, while the negative experiences were mostly attributed to degradation in PI, e.g. the awareness of the VR hardware, or sensory mismatches leading to cyber-sickness.

2.5.2 Methodology

Methodologically, this study joins the work of Slater et al. (2023) in running a purely qualitative assessment with verbal interviews that were analysed through Nvivo 11, a readymade tool to perform qualitative analyses of recorded interviews.

The strength of this study, compared to that of Slater et.al. is the sample size, which, with its count of 76, is almost three times larger than their 26, as described above, meaning the findings can be described as more authoritative.

2.5.3 Studied Subject

The authors of this study approached the project from a therapeutic point of view, aiming to facilitate psychological assessment, and to demonstrate the viability of VR simulated scenarios for such purposes. Its findings can certainly be used for any immersive simulation, however the finding that the effect was strongest when it elicits "genuine cognitive, emotional, and behavioral responses" is akin to suggesting just 'doing it well'.
The outlined themes and subthemes identified from the interviews can be read like a guide for developers of what to look out for, and feelings or reactions to avoid or promote, which, while not a comprehensive guide of do's and don't's, can still be helpful in improving the starting point for any VR simulation.

2.6 Summary

In short, we have defined the study's title to be evaluating plausibility, the illusion that events in a VR environment are 'real' despite knowledge of the contrary, in an immersive virtual environment featuring a beach volleyball scenario, which will inspect the effect of changes in the density and involvement of a virtual crowd on the side.

In highlighting closely related works, we find that the studies into sports-specific adoption of VR, examplified by Stinson and Bowman (2014), used Psi-adjacent concepts as proof-of-concepts for the viability of VR in the specific scenario, in this case the arousal of anxiety in participants.
Other studies more directly tasking themselves with the analysis of Presence or Psi, such as Riches et al. (2019) or Slater et al. (2023), have in the larger sense made use of qualitative methods to provide extensive insights into the underlying factors affecting this sentiment.
Finally, a study into the realism of crowd sentiment showcased a positive use-case for considering direct participant-controlled, level-based configurations directly in-simulation for assessment, whilst also highlighting useful insights into crowd behaviour and perception.

Having laid a solid foundation of information to understand the aims and proceedings of this project, the next chapter will build on the information outlined here to flesh out the identified research gap, and propose a research question and methodology to attempt to fill this gap.

Study	**Plausibility Measured**	**Methodology Comparison**	**Subject**
Stinson and Bowman (2014)	indirectly, and in limited fashion	physiological measurements + standardised tests	viability of VR for psychological training
Slater et al. (2023)	directly measured	sentiment analysis of essays + small questionnaire	improve on non-sporting simulation
Glémarec et al. (2021)	not measured	direct participant choices	crowd sentiment realism in behaviour and animation
Riches et al. (2019)	indirectly measured as part of Presence	qualitative analysis of semi-structured interviews	viability of VR for therapeutic purposes on mental conditions

Table 2.1: A comparison of the four presented closely-related works, with the ways Psi is measured, a quick overview of the employed methodologies, and the overall subject of study.

Chapter 3

Research Aims and Methodology

Having introduced the subject in a broad sense in chapter 1, and laid out background knowledge and existing works in chapter 2, it is now time to look forward and describe first the aims of this project, by way of identifying the research gap it aims to patch up, and the research question making up the substance of it. Following this, the proposed methodology will be laid out, alongside a description of the various components envisioned to make up a patch to fill the identified gap, that won't fly off under the first bit of scrutiny.

3.1 Research Gap and Research Question

The information laid out over the past two chapters, particularly the second, will be leveraged in order to outline all the edges and corners of the identified literature gap, then quickly outline a research question cut out to fit over it.

3.1.1 Identified Challenges

Chapter 2 established that existing use-cases of VR in high-level sport, as shown for Stinson and Bowman (2014), the question of plausibility was only tangentially addressed for the purpose of the project, which served more as a proof-of-concept than anything else.
With Slater et al. (2023), and Riches et al. (2019), two approaches to measuring and improving the sense of subjective immersion, described as Psi in the former, and sense of presence in the latter, in order to make a simulated environment feel more lifelike, each resorting to detailed participant feedback in the form of essays or semi-structured interviews to extrapolate what should be done to improve the experience, each study leading to their own recommendations for best practises in the investigated scenarios.
Additionally, Glémarec et al. (2021), whose study served to flesh out a more expressive crowd in VR, illustrates, not only that certain features of body language pair in various

ways to exhibit widely different sentiments, but also that it neither takes a full body, nor even a full crowd, to impress these sentiments onto participants.
And finally, in the introduction (Chapter 1), the lack of meaningful such applications for use in volleyball was mentioned as well.

This absence is probably not down to chance, although there is no scientific evidence to back this up directly, considering that adapting volleyball to VR confronts developers with many limitations of the technology in development, as they are outlined in Gerwann et al. (2025)'s review, including but not limited to:

- the lack of haptic feedback to e.g. improve setting, which would be difficult to translate to real play at competitive levels,
- a reduced FoV which can lead to important peripheral information escaping perception,
- the limited ability to receive very precise physical cues in a timely manner, examplified in the above review how batters in baseball need to rely on the pitcher's kinematics to predict ball speed, rotation and curve before it goes flying,
- the sheer complexity and granularity of some movements, in particular the set that would either need expensive programming to map accurately, paired with robust hand-tracking technology for use on professional athletes,
- the use of complex, and potentially hazardous movements in smaller spaces, e.g. diving for the ball in a controlled manner, or a very specific run up to jump when spiking in order to maximise jump height.

In this same review, the author emphasizes through these limitations that VR will never be, and never should be, a full substitution of training in physical environments, and should only be used where it makes sense to use for a real benefit. As a follow-up, the author explains that the potential for VR is highest towards injury prevention and the sharpening of mental skills, e.g. subjecting athletes to matchday pressures, or particularly stressful situations such as goalkeeping training under penalty shootout conditions in football/soccer Gerwann et al. (2025).

The aim of this research project being to contribute towards adoption of VR to enhance volleyball training regimes, it is the hope of this researcher to shorten development and design time for such training applications and scenarios with projects such as this one, by proposing best practises to form better starting points than guesswork on developers'

sides.
Considering that crowds form a large part of pressure during matches, it is proposed to look into crowds more specifically, in order to find out in what ways virtual crowds may interact with participants, to be able to suggest potential best-practise configurations to support Psi during gameplay, and, by extension, to keep motivation and performance up while immersed in VR training scenarios.

3.1.2 Proposed Work

As such, the question of what the study must investigate amounts to:

- assess participants' level of Psi during gameplay,
- determine the level of comfort and proficiency with the volleyball mechanics, to mention it as a potential factor in plausibility breaks.,
- provide different levels of crowd involvement to provide viable best practise estimations.

All of this subtly builds up to the research question this paper poses:

> **"Do trade-offs in environment population, such as variations in crowd population affect the illusion of plausibility in a VR Beach Volleyball setting?"**

It is a question that may be tentatively answered by a few guiding hypotheses, such as

- The VR Beach Volleyball game is suitable as a baseline to establish a state of presence, including both PI and Psi.
- Users are fundamentally affected by the crowd in one way or another - i.e. the crowds are plausible.
- The digital crowds can reach a level of activity considered comfortable for users to play with.

Furthermore, a side hypothesis of "Users are more comfortable with a crowd than without one" can be proposed as well. In order to verify these hypotheses, a methodology is proposed in the following section.

3.2 Research Design

In order to look into the established research question and accompanying hypotheses, and considering it involves insights into participants' thinking processes, the experiment will consist of a task the recruited volunteers will be asked to perform, in order to gather data and an opinion to be evaluated towards establishing key findings.
In this section, we will first lay out a rough description of this above task, particularly how it will relate to participants' contribution towards research data, as the technical implementation will be outlined in much more detail in Chapter 4. Following the task description, a short overview of the hardware used for experiments will be presented before introducing the participants recruited for this study, as well as the procedure undergone for every individual trial.

3.2.1 Task

The task for the application was to play in a beach volleyball setting in VR. As soon as the first ball, always prepared for a serve, is played, participants are allowed to play a theoretically unlimited amount of points, regenerating after the previous ball is 'killed', i.e. hits an obstacle, or exits the bounds of the simulation space.
Every point starts with either a serve, or a ball played in from the other side of the court towards the player. There is a 1/3 chance to serve, and a 2/3 chance of a ball being played in, determined using a pseudo-random number generator for every point.

The methodology for the task is inspired by a recommendation outlined in Murcia-Lopez et al. (2020), to let participants influence the environment to their preference instead of either imposing or suggesting a qualitative ranking along the lines of 'configuration A is better than B' without a valid reason to do so. This is motivated by the finding that a more detailed configuration may lead to a decrease in plausibility due to e.g. failing expectations paired with the increased realism.
Therefore, the main variable given to participants to adjust was chosen to be crowd density. Four different levels were provided, increasing in count with rising density. Of a full, generated stand, each level $n \in [0, 3]$ showed every n^{th} participant out of 3, i.e. with a crowd filling of 0%, 33%, 67%, and 100% respectively. The crowds also produced a different audio track for every level, that increased in perceived crowd size with every level.

Participants enter the VR simulation on the highest setting, to get used to the environment, and start play in a 'free play' mode, in order to get used to VR and gameplay mechanics. During free play, the crowd always stayed at level 3, and points were not

counted.
Once they exit free play, they are encouraged to first explore all the levels, then adjust the crowd to their preference afterwards. It was possible for participants to stay at a level they liked for as long as they liked. Participants were also asked to end the experiment on the crowd configuration they preferred, all things considered. A prompt to adjust the crowd, in form of a UI field at eye level, appeared after every 5th point recorded, and only showed the change in crowd after the next level was selected and confirmed.

Points were counted and displayed in the environment to provide an incentive to perform well, although the fact performance was neither recorded nor rewarded was made clear to participants. This means that points served mainly as a stimulant for more competitive users, and a performance feedback mechanism for everyone else.
A history of crowd states is recorded for every participant, and will be used in the results discussion to outline findings.

In total, participants are asked to play at least 20 minutes including free play, in order to produce a large enough sample of transitions.

3.2.2 Hardware

All participants completed their trial using a standalone Meta Quest 3 device provided by the researcher, featuring a resolution of 2064x2208 pixels per eye, with a horizontal FoV of 103.8°, and a refresh rate of 120 Hz. The headset was complemented with a pair of Meta Quest Touch Plus controllers.
The study was conducted exclusively in the SCSS Computer Labs at Trinity College Dublin, in enclosed rooms rearranged to maximise the space available to participants, which gave approximately 3.5m by 4m ($\sim 14m^2$) of unobstructed space to participants. This was made to allow some flexibility of movement, notably for swinging and turning physically rather than relying solely on digital movement, to ease on potential for cybersickness.

3.2.3 Participants

After receiving approval from the Trinity College Dublin's School of Computer Science and Statistics Ethics Committee, participants were recruited through direct contact with the researcher, either via e-mail, or in-person at the place this study was conducted and written. In total, 14 participants were recruited, and all completed the study.
No compensation was offered to participants for their time, most were graduate students

at Trinity College Dublin, and the only age ranges represented were 18-24 and 25-34, specific ages were not recorded. 3 of the participants were female, and 11 were male. None of the participants had any experience with Volleyball prior to this study, and prior experience in VR ranged from never having used such devices before, to regular users.

3.2.4 Procedure

Upon arrival, each participant was greeted and asked to fill out the Informed Consent Form. After a short explanation of the procedure, participants were offered a bottle of water, and asked to put on the HMD and controllers, with help offered by the researcher for users with little prior experience with VR.

Participants then performed the task outlined above, with the researcher in the room to make private observations on exhibited behaviour, answer questions, and direct participants back towards the centre of the active space in case they moved closer to the surrounding walls. Special care was given not to reveal the full purpose of the study, beyond its title, to prevent participants from overthinking the influence of the crowds on their experience, as recommended by Murcia-Lopez et al. (2020). The researcher never actively engaged participants in conversation during gameplay, only intervening unprompted to direct participants towards a safe middle in the room.
During the gameplay phase, participants were permitted to take a break whenever they wanted or needed, which would pause the 20 minute minimum play time.

Following the experience, participants would cleanly exit the game, then proceed to answer a questionnaire relating to the experience, following which they were thanked for their participation. No study session lasted longer than 60 minutes overall.

3.2.5 Evaluation

The data collected towards this study is recorded in two separate prongs: Firstly, there is the crowd state history established during the participant task, which aims to showcase a trend in transitions. This history will also be used to establish a ranking of crowd states by final preference, due to the instruction to participants to close on their preferred configuration.
Questionnaire answers containing quantitative answers to certain questions on Likert scales, as well as open text answers to complete sections, will be leveraged to evaluate the hypotheses with quantitative answers, whilst also hopefully showcasing interesting

correlations, or in the case of open text answers, illustrate or nuance findings through direct user feedback.

Observations made by the researcher during participants' gameplay, such as particularly salient behaviours will also be used to illustrate or nuance findings, though they cannot be used to directly prove or disprove hypotheses, considering expressed behaviours don't always match inside states and may mirror other factors, e.g. the presence of other people in the space, or comfort in exhanging with the researcher or not.

3.3 Summary

In short, we have outlined the potential uses of a study outlining the effects of crowds on plausibility, to avoid breaks in presence during gameplay in VR sports application, which would lead to a loss in motivation and performance in athletes using such technology to support their training regime.
This potential aims to be addressed by this study by answering the question, whether trade-offs in crowd population meaningfully affects plausibility in a VR beach volleyball scenario, chosen for its relative absence of scrutiny in existing literature.

This question is addressed in a participant study featuring 14 young adults, within the age range of 18-34, without any prior experience in volleyball, playing a purpose-built game in which they would either serve, or react to balls played in from the other side of the net. During gameplay, they would adjust crowd density using a regularly appearing prompt, producing a history of crowd transition ending with a given participant's preferred state overall. Participants also answered a questionnaire to assess their reactions in a more quantitative fashion, with open text fields to relate any nuance or more expressive answers.

Chapter 4

Implementation

In the previous chapter, the task participants were asked to perform in order to measure plausibility of crowds was described. Performing this task, predictably, requires the existence of a platform to do so.
In a first section, the absence of a viable, commercially available product to do so is laid out, before the purpose-built game made towards this project is detailed in follow-up sections, broken down by individual components.

4.1 Absence of a commercially available alternative

For use of a standalone VR device, when looking for applications to download and use, the Meta app store is the main platform to use. A search was made to find games with the only keyword queried being 'volleyball', which produced only four results: "Volleyball Fever" Neuston AB (2021), "Volleggball" Pink Marmot Studios (2021), "BIG BALLERS '26" Doge Labs Inc. (2023), and "Clawball" ARVORE Immersive Games Inc. (2024).
The last two are games for a different sport altogether, tailored for basketball, and football respectively. Of the other two games, neither featured any visual crowds whatsoever, and the trailer of Volleyball Fever did not indicate any audio tracks to represent a disembodied crowd, while Volleggball includes these when points are scored.

This researcher neither had the skills nor the time to attempt to mod either of these games for the purposes of this project, therefore the decision was taken to make a volleyball game from scratch using a conventional game engine.

4.2 Overview of the Solution

The requirements for a game supporting the participants' task are for the game to take place in a suitable visual environment, enable classic interactions with the ball as accurately as possible to resemble their real counterparts, a baseline set of volleyball rules, and to feature a crowd that could be dynamically adjustable by way of a user interface (UI) field.
In order to accommodate the free play mode, there was also a need for a baseline tutorial phase, that also did not count points or propose the crowd adjustment panel. Finally, to increase engagement, a varied set of play situations was considered desirable.

For development, two game engines were considered for their status as industry-leading, as well as the researcher's prior experience in working with both. Unity (Unity Technologies (2026c)) was chosen over Unreal Engine (Epic Games Inc. (2026)) due to the researcher's comfort in using the engine in prior projects, alongside Unity's comprehensive VR template packages that already implemented many VR features, such as grabbing and throwing, or the provision of a fully VR-rigged avatar with basic movement and interactions. The chosen version reflected the latest update at the time development started, Unity 6000.4.8f1.

Over the next few pages, the various components of the projects will be outlined in order of implementation, beginning with the template assets, followed by the crowds and game rules, then a quick primer on Unity's physics engine before going into ball physics and interactions, then closing off with any parts of components that didn't deserve a dedicated section.

4.2.1 Unity's VR[MP] template and Scene Setup

This project leverages Unity's Virtual Reality MultiPlayer (VRMP) package, which leverages built-in implementations of OpenXR, an API enabling the use of input from various types of VR hardware, as well as Unity's XR Plugin Management package, to provide a wide range of features and prefabs - ready made game objects developers can use immediately without dependency - that are fundamental to any VR application.
This package specifically provides a full implementation of movement, both continuous along the ground using a directional joystick on the left controller, and teleportation to a point aimed from the right controller whilst holding a focused button. Including both modes of transport is advantageous, for the former's ability to navigate all reachable terrain, whilst teleportation is an effective mode of virtual transportation that notably can

negate the potential for cyber-sickness, albeit at a relative loss in presence Caputo et al. (2023).
A functional virtual jumping motion is also included in the package, although participants mentioning a history of cyber-sickness were discouraged from using this feature.
From the packaged prefabs, this project also leverages the implementation of grab interactables, enabling users to grab the volleyball for serving, as well as the visualisation of a ray for UI interactions. Finally, the avatar and linkage of controllers were imported from the template's XR Rig prefab, which included the targeting of the virtual camera to the players' display, and the tracking of controller position and rotation within the scene as well.

Regarding the visual environment, the raw terrain was generated procedurally in and exported from Blender, following a Youtube tutorial by ChuckCG (2025) for the base setup. The animated water texture as well as beach decoration assets were sourced from the "Tropical Beach URP" Sullivan (2022) package, whilst the volleyball-specific assets, i.e. the net and ball, alongside associated textures, were sourced from the "Volleyball Set 1" Listudio (2024) package, both purchased from the Unity Asset Store.

4.2.2 Stadium Crowd Generator

Having set up an appealing scene, the stakes for the crowd were high. VR applications are required to run at a higher framerate in order to avoid cyber-sickness, where 60Hz would already be considered too low to be viable, and showing optimal levels starting at 120Hz with consistent improvement in-between (Wang et al. (2023)). Therefore the crowds needed to be performant enough to not significantly impact frame duration, whilst also allowing for dynamic adjustment.

Against all hope, the Unity Asset store actually produced a product that checked all of the boxes, with the "Procedural Crowd Generator" plugin (AiKodex (2021)). This plugin builds - prior to runtime, so the procedural aspect doesn't eat away at loading time - a stand featuring a custom amount of rows and columns of spectators, varying between 7 variants of humanoid spectator prefabs with a decent diversity in both gender and ethnicity, as showcased in figure 4.1. And importantly, every part of the plugin, i.e. meshes and animations, came with performance optimisation in mind. After running a few tests, the game featuring two separate stands of 8x22 spectators ran without a dent in performance, earning its spot in the project, sparing a lot of effort that would otherwise

Figure 4.1: A screenshot of one of the stands, featuring a diversity of spectators randomly placed in.

have been required to reach any level close to this.

The plugin was not, however, entirely perfect for the purpose, prompting a few small adjustments that were thankfully not a mess to add, as the source code running the plugin came with the package. The first adjustment was randomising the generated seating, as the plugin would simply rotate through the spectator variations when assigning each character. Instead, for every seat, a pseudo-randomly generated number would determine which prefab was chosen.
The second amendment to the crowd plugin was to dynamically adjust crowd density. The initial hope was that the built-in "distribution" setting, changeable in-editor between "Standard", "Easy", and "Compact", at rising levels of density, could be leveraged directly, with the addition of an 'empty' setting that would remove the crowd altogether. Unfortunately, that distribution did not trigger a re-generation of the crowd when adjusted during runtime. Consequently, the spectator prefabs were assigned a specific tag, which would enable a new CrowdManager script to reference all spectators when starting play, to then disable the desired proportion of characters when changing the crowd density for the purpose of the research study.

The added, small benefit of the crowd generation being done during development is that all participants experienced the exact same crowd for every game, signifying the elimination of any potential bias from the proximity of different spectators between participants.

The final touch to participants was to add an audio track, which did not come with the plugin. The goal was to, for every crowd level featuring at least one participant, gather at least two different audio tracks for the overall background, that would loop, and one audio track for cheering whenever a point was scored.
Audio tracks were sourced from pixabay.com from sounds marked as royalty-free.

4.2.3 Physics and Ball Properties

After having set up the environment, as well as a controllable crowd the next item of interest is, of course, the ball.

The implementation itself is limited to adding the ball prefab containing the mesh and texture into the scene, then adding a sphere collider and rigidbody to give it a presence in Unity's Physics engine, with a few modifications to the corresponding Physics Material for more realistic behaviour.

In order to explain these, a quick exposé on Unity's physics system (Unity Technologies (2026a) is due.

For every object desired to have a physical presence in the virtual environment, there are two components to consider: the Collider, which represents the shape and [physical] presence type of the object, and the Rigidbody representing the physical properties of the object, i.e. mass, the use of gravity, or linear and angular damping to simulate friction.
Colliders can separate the type of collision between two objects: if a collider is marked as a trigger, it does not need a Rigidbody component, and will not trigger physical collisions with objects with a Rigidbody. Instead, they will produce OnTrigger events, which are mostly used to detect objects passing in a certain area.
If a collider is not marked a trigger, it should be paired with a Rigidbody to simulate physical collisions, and will trigger OnCollision events when colliding with another physics body. The rigidbody can disable gravity, and can also be designated as 'kinematic', which will exempt its object from external influence. Any impulse, collision, or force applied to it is simply discarded, and for all intents and purposes, the object is unmovable through the physics engine, although its position and rotation can be manually overwritten using its Transform component.

The PhysicsWorld defines core and default values for e.g. gravity calculations, which default to natural values. In the case of Gravity in 3D, it is defined as a 3D-vector defaulting to (0, -9.8, 0), representing a pull of 9.8N straight downwards, applied to every object with a Rigidbody component with gravity enabled.

When they do not collide, objects with rigidbodies move through the environment based on their velocity, a 3D-vector representing their movement speed from one frame to the next in units per second. This velocity is adjusted to reflect the influence of gravity and simulated friction, but the aftermath of collisions with other bodies.
Inorganically, developers can also affect a rigidbody's flight path by adding raw impulses or forces through code, which are internally updated by unity algorithms based on the type of force applied. Velocity can also simply be overwritten, though that is often undesirable due to it producing unnatural behaviour.

The collision solvers run at different settings depending on required accuracy and granularity, defaulting to the lightest of configurations, the discrete solver which only considers the position in a given frame. The issue with discrete collision solving is, that objects travelling fast enough to theoretically pass through a given object between two frames would not trigger any collisions, and therefore lead to faulty behaviour. The alternative settings, "continuous", "continuous dynamic", and "continuous speculative" increase the accuracy of collision detection, costing performance as compensation. Due to the low amount of objects requiring collision detection - the ball, ground, and invisible boundaries - and the absence of performance issues, the most expensive solver was tested, and adopted after playtests did not result in noticeable performance drops.

Now armed with a grasp of Unity's physics simulation system, this last paragraph will outline the material properties governing the ball's physics behaviour.
Mirroring a volleyball's average mass of approximately 230g, the ball's weight is set to 0.23 mirroring Unity's default weight unit representing kg. Linear and angular damping, or friction in the air, were left at the default values of 0.1 (Unity doesn't provide a real unit in its documentation Unity Technologies (2026b) for comparison, and flight behaviour looked realistic enough), and the material properties were mostly adjusted for behaviour interacting with the ground, as hand collisions were later discarded as will be explained in section 4.2.5, and material properties associated with the ground didn't seem to propagate to the ball in first tests. Static and dynamic friction, representing resistance to sliding and spinning motion respectively were left at the default values of 0.6, while bounciness was lowered to 0.2 - or 20% of potential on impact - to mirror and slightly exaggerate the reduced, but not nonexistent bounce on sand.

4.2.4 Game Rules

Point Recognition:

The fundamental game rule of volleyball is point recognition, i.e. recognising when an object collides with an obstacle not considered part of the play. For this purpose, the ground was beset with a kinematic collider (not impacted by direct physics influence), tagged as 'Ground' for separation from other colliders. To recognise whether the ball landed inside, or outside the court, a pair of transforms was placed on the corners of the playing field - at the outside of the blue line, which was aligned so its edges went on straight in x and z-direction. i.e. the plane could be represented using a point P at any corner as $P + a * (1, 0, 0) + b * (0, 0, 1)$, with a, b lower or equal in magnitude than the length of a given side.
With the recognition points at diagonally opposed corners of the pitch, this gave a pair of boundaries $(x_{min}, z_{min}), \quad (x_{max}, z_{max})$. If the collision point between the ball and the

ground is within these bounds, i.e. $x_{min} \leq x \leq x_{max} \land z_{min} \leq z \leq z_{max}$, then the ball is considered in bounds, otherwise it is out of bounds. The y coordinates representing elevation are not relevant for point handling.

If the ball is in bounds, however, it is also important to determine which side of the court it landed on. Given the court stretched in z-direction, the net's z-position was noted, and used as a discriminator. Given the net was befitted with a collider, it was technically impossible for the ball to land exactly on that z coordinate, so if $z > z_{net}$, the ball landed on courtside A, assuming A was on the side of the net with greater z values - the discriminator was set using a bool value determining whether Team 1 was occupying courtside A or not. This simple comparison-based point recognition system is very lightweight and practical, compared to relying on multiple, potentially conflicting colliders for different pitch sides.

The ball was given a boolean parameter `LastTouch`, which stored whether the player's team - always Team 1 on the scoreboard - touched the ball last, or not. If the ball landed out of bounds, the point is given to the team that did not touch the ball last, and if it was in play, Team 1 gets the point if it last touched a ball landing in the opposing team's half, otherwise the point went to Team 2.

World Boundaries:

No virtual world is infinite, given the existence of some level of hardware and computational limits for every possible device on the planet. It is therefore good practise to implement some movement barriers the player cannot bypass, to avoid the ultimate punishment of falling through the environment, should they reach a place without any solid ground to navigate on.

The simplest solution which requires no budget at all, and comparatively little implementation time, is to add physical walls that do not render visually, only existing with a presence in the Physics System.

Therefore, an invisible barrier was set around the perimeter of the stadium, barring the player from going into the stands, and further towards the shore than the scoreboards above the palm trees. This barrier also blocks the ball, which resulted in collisions with this barrier being considered a kill of the ball as well, as they would represent an obstacle that prevents the ball from flying out indefinitely.

The scenario of the ball flying out indefinitely could still kick in on a misunderstanding, as a point would, until now, only be granted if a collision with an object tagged as

'Ground' is recorded. To patch this scenario, a trigger collider encompassing the stadium was fitted, where upon exiting this bounding box, the ball would also be considered out of bounds and killed, as it was unsalvageable.

Ball Lifetime:

After a point ends, a short waiting period is set, whereafter the previous ball is destroyed, to reappear on the pitch, ready to be played. For every point except the very first, which will always be a serve, a pseudo-random number generator rolls a dice - random integer in the range $[0, 3[$ to determine the type of start to the next rally. If the number is larger than 0 (or equal to either 1 or 2), the ball would simulate being played in by the opposing team, otherwise it would be placed at a serving position for the player to grab at their leisure.
If the ball is played in, a notification reading "Incoming!" will be shown to the player, and the ball appears at a random point above the net, and is calculated to aim above the player's head, at a speed calculated so that it will reach the player within a random timespan $T \in [1.2, 1.5]$ seconds.

This was done by establishing the x-z direction as a simple vector from the ball to the player, then calculating the y velocity making up the ball's curve, in order to reach the target position in time T. This calculation for the y velocity was ideated with assistance of the AI agent Claude to speed up the establishment of a viable function. The function reads as follows, with g representing the gravity in y-direction in Unity's Physics Engine, and $\overrightarrow{dir}$ the 3D vector from the ball to the player ($\overrightarrow{dir} = \overrightarrow{player} - \overrightarrow{ball}$):

$$y_{vel} = dir.y - \frac{g * T^2}{2}$$

, citing a kinematic equation used in fundamental ballistics, $s = \frac{gt^2}{2}$, attributed to Galileo Galilei.
The overall resulting vector $\overrightarrow{v} = (dir.x, y_{vel}, dir.z)$ was then divided by T to add in the time constraint. This meant that the ball would travel faster the further the player was from the ball's position, encouraging players to move further up field for these rallies.

If the ball is presented for a serve, the ball is made grabbable by activating its XRGrabInteractable component, allowing users to take a hold of it. Once the ball is released, the ball is sent upwards by a constant force, in order to mimick a serving throw. Attempts to implement this serving throw as an actual continuation of hand movements resulted in jerky throws that often went out of reach, and away from expected directions due to

the inherent difficulties in estimating throwing directions in VR (Zindulka et al. (2020)). Therefore velocity was straight overwritten to an upwards vector $(0, 1, 0)$, scaled by a serveThrowForce parameter in-editor adjusted through trial and error, yielding a value of 7,5.
The XRGrabInteractable component is disabled on release, to ensure players don't catch the ball during gameplay.

Once the ball is killed, i.e. as soon as it collides with the ground or any obstacle, the ball rolls on naturally for a few seconds, ignoring any collisions with the player, before being destroyed by the game managing script.

Match Events:

During the match, results are stored by a MatchManager script, that tracks the score and crowd state. Matches are set to run to 11 points without a tie breaker, and every five points, the user is presented with a prompt to adjust the crowd. This is specifically checked by adding the scores and running a modulo operation `score[0] + score[1] % 5 == 0`. Once a match is over, the scores are reset and a new rally starts automatically.

4.2.5 Hit Mechanics

Hit Strength

For ball interactions, the first step is to get the collisions with the hand working properly. The initial hope was to be able to leverage Unity's Physics Engine by equipping the in-game hands model with a collider - initially a box-shaped collider for simplicity - and rigidbody that would not take any physics influence from outside, but handle its velocity like normal. The removal of physics influence is a requirement, lest the hand stop mirroring the controller's position.
Unfortunately however, this was not to be, as physics bodies subject to controller-tracking do not adopt or pass on a velocity value implicitly to the physics system, and therefore any collision with the ball would be interpreted by the physics engine as a collision with a static wall for a given frame, losing all the impact of the hands in the process.

The following attempt was to then explicitly pass on the impact through a trigger event instead of making the hands a physical brick, and estimating the aim through baseline measures detailed later. The only problem was how to get the velocity in.
Several attempts were made to gather this information directly in-engine, such as using the raw difference in position between two frames (`velocity = lastPosition -`

`transform.position`). This led to many issues including lack of consistency, and the exclusion of factors like momentum, by which way a jerky reaction would be interpreted the same as a stone cold spike on the moment of impact, or how e.g. braking a move, common to alter a ball's flight path for floating serves for example, would negate velocity altogether.
Various methods of smoothening were attempted in order to remedy these issues, such as weighted, and unweighted averaging over 5 frames, each presented its own problems, and none were reliable. In testing hit strength over several runs, for the same arm movements, the engine would one time record hand speeds in the range of $[0, 2]$ m/s, whilst in the next run, the range would inexplicably rise to $[0, 10]$ m/s.

Finally, with a bit of documentation perused and many fruitless forum discussions read, it was noted that the Input System has a built-in function to read controller velocity in game space, via embedded sensors. It is not a perfect solution, as it also doesn't store momentum as mentioned above, a very important factor in the resulting hit strength, however it was at least consistent over several runs, and was adopted as a baseline due to the rapidly approaching development timelines.

Therefore, hit strength was calculated proportionally to hit speed on impact, moderated by a hit modifier which will be detailed in the next subsection for their specificity to certain hit types, which at least had the benefit of making that factor relatively independent of users' physical strength.

Hit Classification and Aim Estimation

In volleyball, there are many distinct ways to interact with the ball, each leveraging a different and often subtle set of physics and biomechanics to affect the flight path of the ball afterwards. In order to simplify this classification for VR, a model differentiating mainly between two-handed, and one-handed approaches was the minimum, associated with a differentiation between digging and setting, as they involved entirely different mechanics.

One-Handed Hits

One-handed hits were first implemented as a baseline, and never improved upon later as a result of the development phase coming to a close. The aim estimation was set to going in exactly the direction the hand is moving towards when the hit is registered, i.e. the clear kinematic intent.
For the hit modifier, a curve was desired that eases punishment of weaker or slower hits,

for the scenario of e.g. braking a hit, or a more cautious VR user dosing their swings, whilst balancing hits in an esablished mid-range of hit strength, to avoid making the dosing too gimmicky.
The output hit strength should look along the lines of $|f| = g_{1HH}(x) * x$, with x the magnitude of the hand velocity when the hit registers. This would result in a directional hit vector of

$$\overrightarrow{hit} = \overrightarrow{vel_{hand}}.normalized * g(|\overrightarrow{vel_{hand}}|) * |\overrightarrow{vel_{hand}}|$$

Having described the shape of the curve g_{1HH}, which would be a logistic decay function starting at a maximum multiplier MAX, then ending at a minimum multiplier MIN with a steepness k, and a midway point $MID \rightarrow g_{1HH}(MID) = MIN + \frac{MAX-MIN}{2}$ for better control, a function designed with corrections from Claude AI was established with the function

$$g_{1HH}(x) = \frac{MAX - MIN}{1 + e^{k*(x-MID)}} + MIN$$

, visualised in figure 4.2.

The hit strength modifier was tested on serving scenarios, both for overhand and underhand hits. After some trial and error, the values converged around $MAX = 55, MIN = 13, k = 0.33$, and $MID = 5, 5$. This had the unfortunate side effect of never applying the MAX modifier outright, however attempts at making this happen through a steeper curve, increasing k, resulted in disproportionately strong hits before the intended mid-range, and so this curve (fig. 4.2) was deemed acceptable and adopted.

Setting

The first difficulty in implementing this mechanic was the differentiation between one-handed hits and the two-handed ones. This was made by storing "hit data" collecting hit position, hand speed on impact and other information necessary for aim estimation separately from each hand, and erasing hit data when the hit is 'processed' in the next frame. There were initial concerns about the frame delay, however the delay proved both small enough to be barely noticeable in play, and a same-frame solution was hard to conceive due to the sequential nature of programs in general, otherwise involving manual estimations for collisions or other metrics of proximity which could have led to their own flaws.
Therefore, if, one frame after any collision is detected, a second collision is also recorded, the hit would be classified as two-handed, and otherwise processed as a one-handed hit as showcased above.

Hit strength for setting was established as the combined hand speed on impact, i.e.

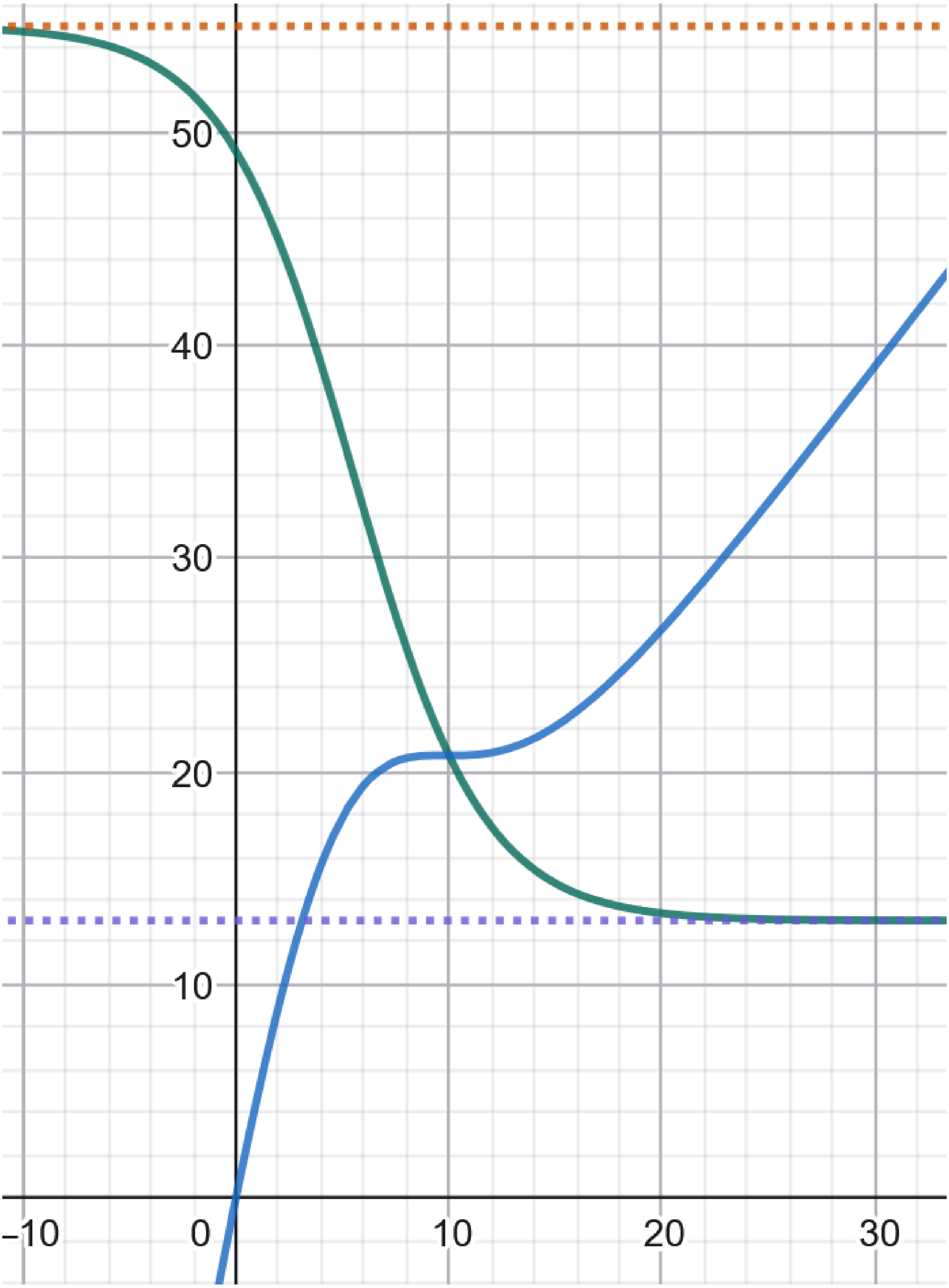


Figure 4.2: A graph showcasing the behaviour of $g_{1HH}(x)$ (green), and the output hit strength $g(x)_{1HH}{*}x$ (blue) divided by 10 for visualisation. The orange dotted line represents the MAX value (55), the purple dotted line represents the MIN value (13). $k = 0.33$, and MID is set to 5,5.

$|vel_{LHand}| + |vel_{RHand}|$. This was again implemented more as a baseline than an accurate guess at setting strength, as the impact of e.g. the thumb, as well as angular velocity of the hands influences the release speed normally.
The strength modifier was here just made to be a regular decay function relying on three parameters MIN and MAX representing the same min and max modifier values as in g_{1HH}, and $SPDMAX$, the speed value after which $g_{set}(x)$ should converge towards MIN, defined as

$$g_{set}(x) = (MAX - MIN) * e^{-b*x} + MIN,$$

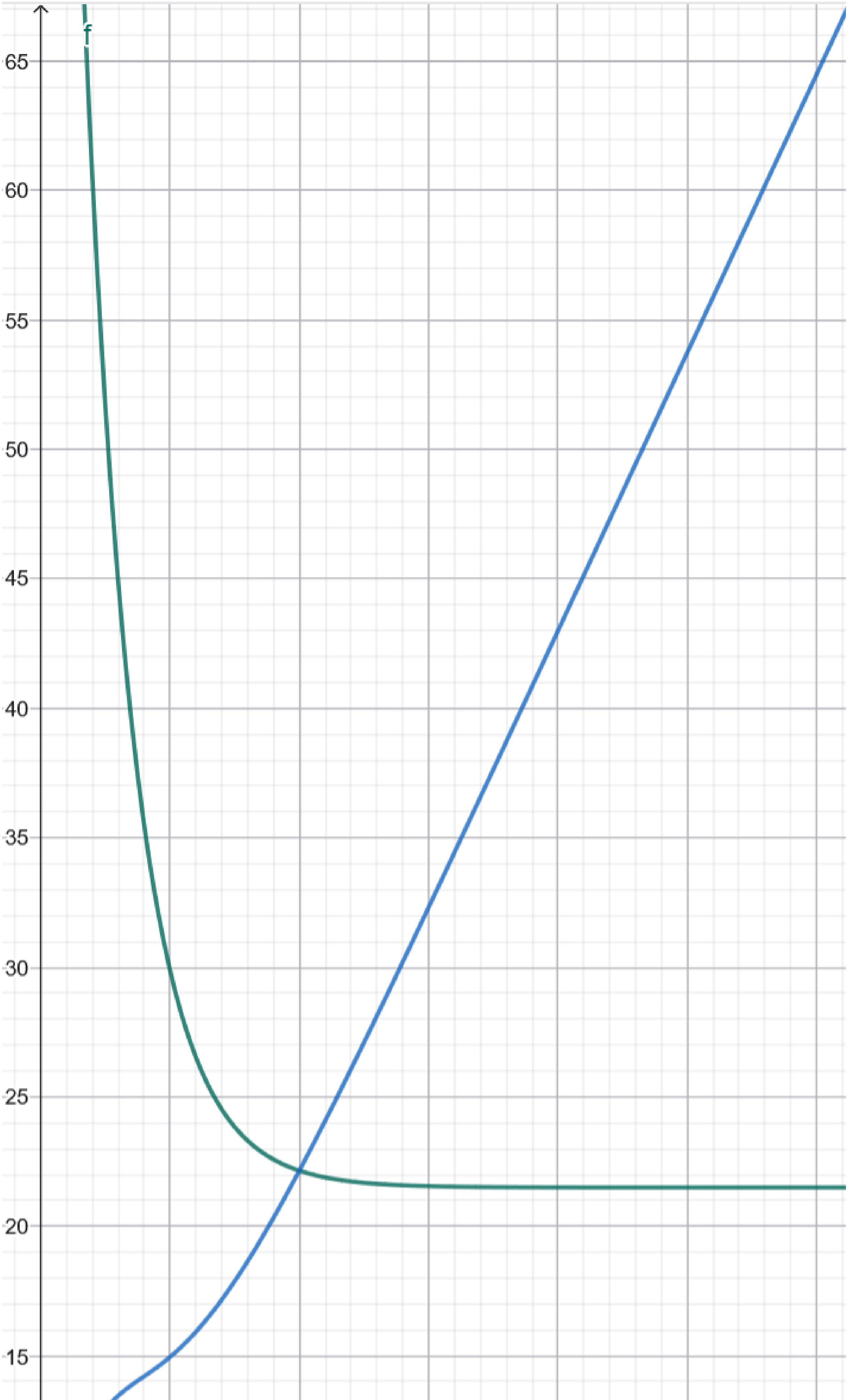
with $b = -\frac{ln(\epsilon)-ln(MAX-MIN)}{SPDMAX}$ and ϵ greater but very close to 0, b having been suggested by Claude to ease on the curve's converging factor, rather than trying to add another steepness factor to adjust on top.
Through trial and error, again testing on serving scenarios, the values of $MAX = 130, MIN = 21.5,$ and $SPDMAX = 9$ were adopted for use.

Aim estimation was more tricky, as it is a complex move to direct even in real life. To efficiently calculate aim, a trick of sorts is often taught to beginners to form a sort of diamond between the thumbs and index fingers of the two, outstretched hands, oriented towards where the ball is desired to go. This could, in turn for engine systems, be simplified to a "hand normal" that would stretch out of each palm, that could subsequently be summed up to establish a direction, as visualised in figure 4.4.
Testing showed it worked relatively well when used as designed, so this aim mechanism was kept.

Digging

The final move to implement was digging. In real life, digs are used to cushion incoming fast balls to bounce off the arms placed together to form a makeshift platform, for the setters to have an easier time controlling the second ball.
Therefore the output hit strength had to be proportional to the speed of the ball on impact, whilst still retaining some proportion of user control to handle e.g. slower balls, or to revive a ball by sending it away even stronger.

Implementing the forearms usually used for this instead of the hands would have involved a time-consuming process of rigging relative to the controller, and more adjustments to not otherwise interfere with the player, so this idea was replaced by an esimation for the platform using the hands. When replicating the digging motion in-game, the hands would align, and it was noticed the thumbs could be an intuitive visualisation of the platform.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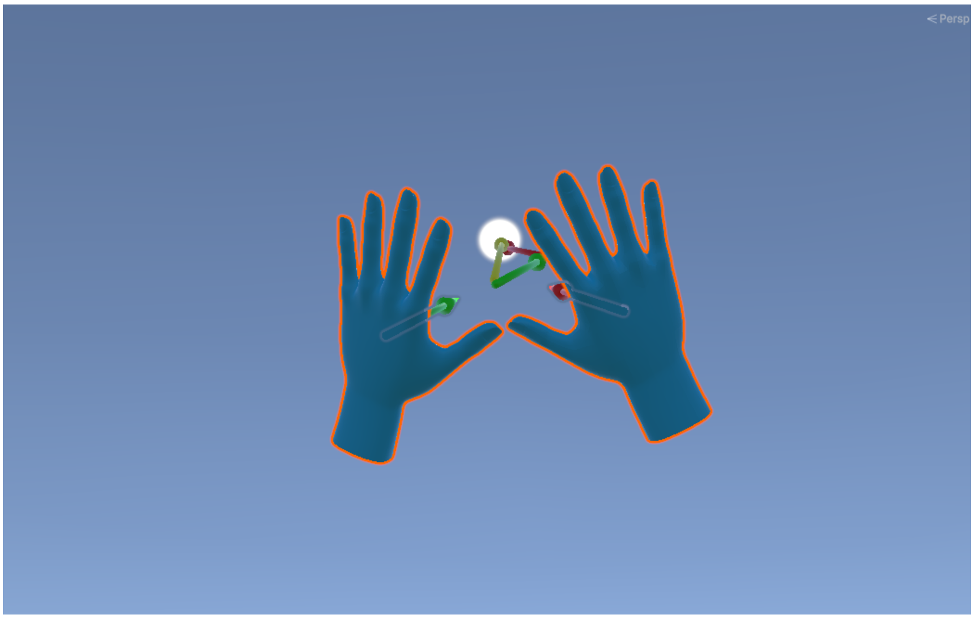


Figure 4.4: A visualisation of the set's aim estimation mechanism, showcasing the hand normals in green and red, and the produced aim direction in yellow.

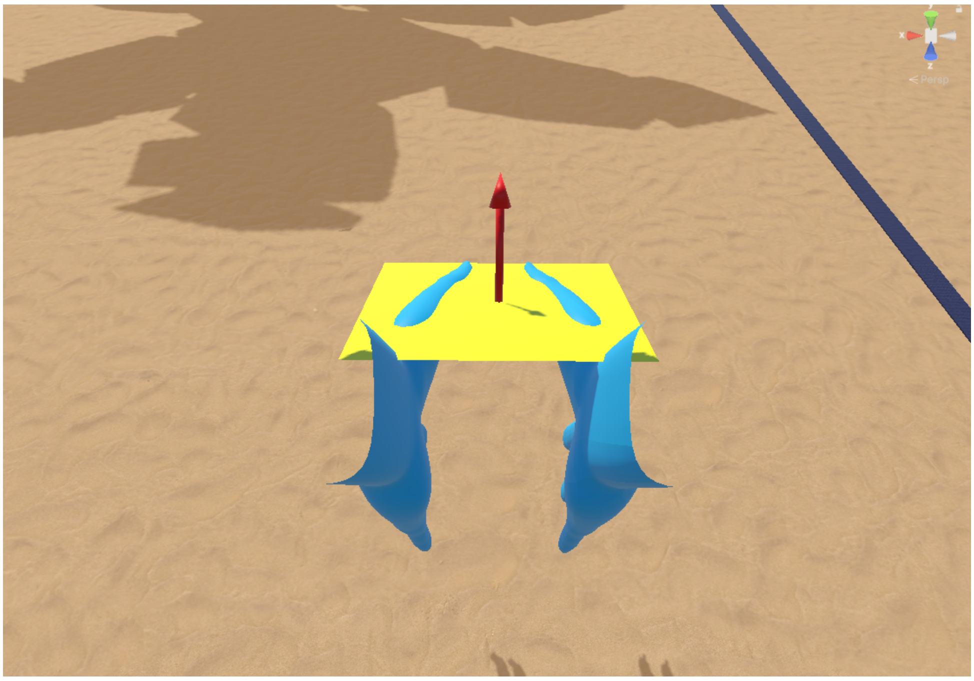


Figure 4.5: An illustration the dig's aim estimation mechanic, estimating a platform between the thumbs, and bouncing the ball off the normal of this platform.

Hands then calculated a "dig normal" N representing the cross product of the thumb direction ($\overrightarrow{L}$ or $\overrightarrow{R}$) and a vector between the palms of both hands (I from right to left) - a sketch to help illustrate this dig normal is provided in figure 4.6 - combined to calculate the normal $\overrightarrow{N} = \overrightarrow{R} \times \overrightarrow{I} = -\overrightarrow{I} \times \overrightarrow{L}$. The 'aim' direction is then the ball's velocity vector reflected across this dig normal using the built-in `Vector3.Reflect` function implemented by Unity.

The speed modifiers for hand and ball speed were implemented as direct multipliers, adjusted through trial and error to the values $mod_{ball} = 20$, and $mod_{hands} = 10$. The two hit modifier were applied in additive fashion, to avoid the absence of one, e.g. a static hand on receive, to penalise the other modifier. The code to determine the dig's aim and direction is provided below:

```
Vector3 reflectDirection = Vector3.Reflect(BallVelocity.normal-
        ized, DigNormal.normalized);
float ballVelocityModifier = ballVelocityMod * BallVelocity.
        magnitude;
float handVelocityModifier = handVelocityMod * HandVelocity.
        magnitude;
Vector3 force = (ballVelocityModifier + handVelocityModifier) *
        reflectDirection;
```

Finally, in order to tell sets apart from digs, a height discriminator was put in place. If the ball was at eye level or above, the move would be classified a set, otherwise it would be interpreted as a dig.
There are certainly better ways to tell the two moves apart, but as a baseline this worked.

4.2.6 Performance

Having implemented everything, a performance assessment was made over one longer playthrough, which yielded an almost perfectly stable 72Hz framerate - please note that this value was a hardware-imposed cap, as opposed to a hard max in performance - whilst utilising 22% of CPU, and 6.5% of memory requirements on the standalone VR HMD described in section 3.2.2.

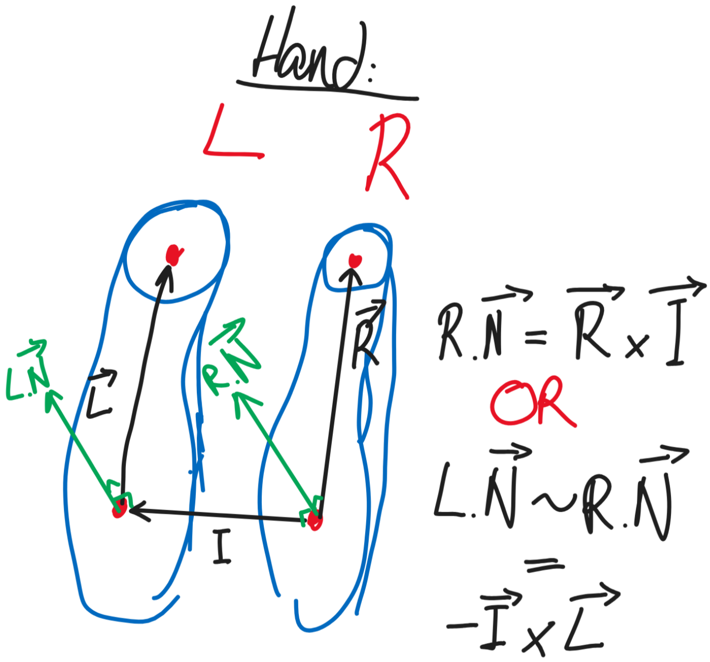


Figure 4.6: A sketch illustrating the vector math used to calculate the dig normal. The order of operations in the cross-product is flipped to accommodate for the right hand rule determining the orientation of the cross vector.

4.3 Summary

In the opening pages of this chapter, the need to develop a purpose-built game for this project was detailed, before enumerating the various requirements, then components of this game.
The starting points provided by Unity's VRMP template package, alongside some assets purchased and downloaded from the Unity Asset Store were leveraged as a starting point for scene setup, and to gain much time and performance for the crowds, despite some very minor retouches.
After outlining the physics system and the properties given to the ball, the implementation by hand of game rules and events, then of various hit mechanics was outlined in all their sensibilities, before showcasing its very good performance overall in the closing stages.

Having now all the background and technical insights needed, the experiment's proceedings will be outlined in the next chapter, alongside the findings resulting from the experiment.

Chapter 5

Evaluation

The Evaluation chapter should present a comparison of the work that forms the basis of the dissertation and existing work. At a higher level, it should demonstrate an awareness of the relationship of the dissertation work to the research area that it is based in.

5.1 Experiments

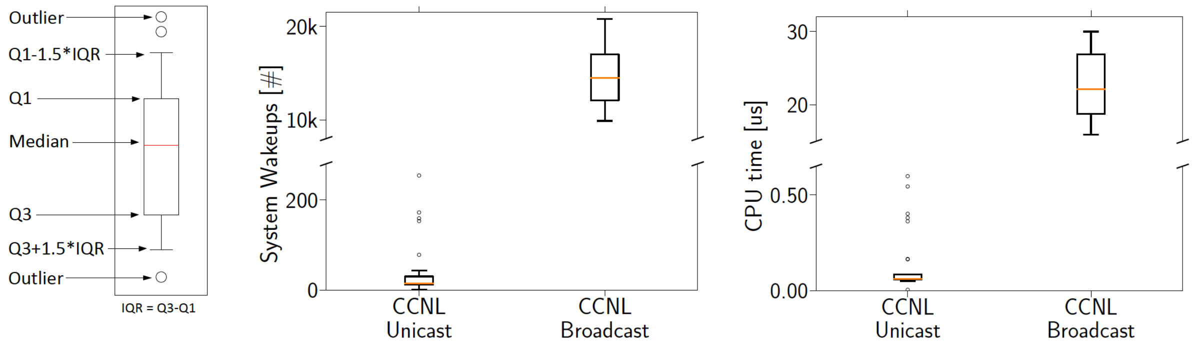
In the case where experiments have been carried out, the experimental setup and the values that were defined for the variables need to be presented in a table e.g. table 5.1.

Column 1	**Column 2**	**Column 3**
Row 1	Item 1	Item 2
Row 2	Item 1	Item 2
Row 3	Item 1	Item 2
Row 4	Item 1	Item 2

Table 5.1: Caption that explains the table to the reader

5.2 Results

Figures that present results such as figure 5.1 need to display descriptions of the axes, the units and scales of the measurements, statistical values, etc. Where measurements were taken from experiments, error bars or confidence intervals need to be provided to give the reader an indication of the spread of the measurements.



(b) System wakeups (c) CPU usage

Figure 5.1: Long caption that describes the figure to the reader

5.3 Summary

Every chapter aside from the first and last chapter should conclude with a summary that presents the outcome of the chapter in a short, accessible form.

Chapter 6

Conclusions & Future Work

This chapter should summarize the work presented in the dissertation and discuss the conclusions that can be drawn from the work and the results presented in chapter 5.

6.1 Limitations

- VB in VR exposes MANY limitations of iVR in general:
  - complete absence of haptic feedback
  - dangerosity from lack of vision of feet/ground
  - need for micro-detail recognition (ex. for blockers, any indication, e.g. eye or head movement, hand orientation and signals, from the setter where the ball will go counts)
  - very complex ball handling compared to other sports (uniquely so)
  - complexity of implementation to a realistic level (many nitpicky rules e.g. double touching, holding the ball for too long during a set, setting foot on the other side of the net by accident etc.)
- simplicity of implementation + faulty aim estimation & hit classification
- questionnaire not nearly good enough: Open text fields not leveraged at all, crowd influence should have been measured from negative influence to positive with neutral/unaffected as a mid value instead of extreme
- usefulness of transition matrix? Yet to look at it more closely but on face value it seems like a bit of a pointless metric idk

6.2 Future Work

The section may present a list of items that were beyond the scope of the dissertation.

Game improvements:

- use of controller-free controls (e.g. hand-tracking, maybe full-body or foot-tracking for more holistic games) - e.g. for specialised systems through a real-time MoCap setup → could be paired with haptic gloves to simulate ball physics on hands!
- Better hit classification (esp. from one hand), refining aim recognition for specific mechanics (e.g. dig particularly)
- including actual opponents rather than just reacting to balls flying in

Study direct improvements:

- use as a learning experience, e.g. using this to teach complex IRL mechanics (e.g. setting) and measuring how well that translates to improvement on the pitch.
- more in-depth probing for results (e.g. sentiment analysis on interviews or similar, more open responses, as lauded by Slater in Slater et al. (2023) to pick up on a much larger depth of information than a 10 point scale ever could)

Tangential directions for further research:

- More alive/reactive crowds (with more accurate levels of engagement, maybe recorded from events of the actual sport; also more individual shouts, more reactiveness to on-pitch happenings (oohs and ahhs with bigger events e.g. blocks), or wider range of emotion - cheering, booing, jeering, chants, boredom/relative quiet etc.), e.g. using researched crowd behaviour models like the one used for Glémarec et al. (2021).
- more detailed investigation into human factors (what kind of trait reacts the most to the taunting spectator, what kind of trait is associated with complete indifference from the crowd etc.)
- add more factors to crowd for participant control, e.g. detach audio from crowd level (what if lv 2 audio was more accurate with a fuller crowd for example).

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Appendix

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